As a side note, the prior theorem can fail when we allow M and N to have nonempty boundaries. For an obvious example of this, let $M = H^n$ and $N = \mathbb{R}^n$. Then let $\iota : H^n \hookrightarrow \mathbb{R}^n$ be the inclusion map and consider any $p \in \partial H^n$. We already know from earlier that $\mathrm{d}\iota_p$ is a bijection. That said, the prior theorem holding in this situation would contradict the invariance of domain theorem on *page 1008*. So, no such open sets U_0 and V_0 can exist in this case.

Exercise 4-2: Suppose M is a C^r manifold (without a boundary) and N is a C^r manifold with a boundary. Also let $\mathrm{Int}(M)$ denote the set of interior / non-boundary points of N. Then if $F : N \to M$ is a C^k map (where $k \leq r$), $p \in M$, and $\mathrm{d}F_p$ is bijective, we must have that $F(p) \in \mathrm{Int}(N)$.

Proof:
Once again, the fact that $\mathrm{d}F_p$ is injective means that there exists an integer n with $\dim(M) = n = \dim(N)$. Now let (U, φ) be a smooth chart on M containing p and (V, ψ) be a smooth chart on N with $F(U) \subseteq V$. Since M has no boundary, we know that $\widehat{U} := \varphi(U)$ is open in $\mathbb{R}^n$.

Yet now note that $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is a C^k map from $\widehat{U}$ into $\mathbb{R}$ such that $\mathrm{D}\widehat{F}(\varphi(p))$ is invertible. It follows by the inverse function theorem in my math 140c notes that there are open sets $\widehat{U}_0$ and $\widehat{V}_0$ of $\mathbb{R}^n$ such that $\varphi(p) \in \widehat{U}_0 \subseteq \widehat{U}$ and $\widehat{F}$ is a bijection from $\widehat{U}_0$ to $\widehat{V}_0$. So $(\psi|_{\psi^{-1}(\widehat{V}_0)}, \psi^{-1}(\widehat{V}_0))$ is an interior chart on N containing $F(p)$. ■

7/23/2026

To start off, it's been bugging me for several days that I don't have a way of talking about tangent spaces to corner points of C^r manifolds (where $r < \infty$). So I'm going to bodge a solution. I am not getting this construction from any resources though so I'm sorry if my construction disagrees with a better solution.

Let M be a C^r manifold with a corner point p. Then we define $\mathcal{C}(M,p)$ and $\mathcal{V}_pM$ as on *page 1046*. Also, we let $\mathcal{D}_pM$ denote the space of derivations at p from $C^1(M,\mathbb{R})$ to $\mathbb{R}$. Now for any $\gamma \in \mathcal{C}(M,p)$ we know $D_\gamma : C^1(M,\mathbb{R}) \to \mathbb{R}$ given by $D_\gamma(f) := (f \circ \gamma)'(0)$ is a well-defined derivation at p. Also, we know by the definition of the equivalence classes in $\mathcal{V}_p(M)$ that $[\gamma_1] = [\gamma_2]$ in $\mathcal{V}_p$ if and only if $D_{\gamma_1} = D_{\gamma_2}$.

<u>Lemma:</u> Let $S := \{D_\gamma : [\gamma] \in \mathcal{V}_pM\}$. Then let T_pM be the subspace of $\mathcal{D}_pM$ spanned by S.

(a) If M is an n-dimensional manifold then T_pM is also n-dimensional.

Proof:
Let (U, φ) be a smooth chart on M containing p, and then recall for any $f \in C^1(M,\mathbb{R})$ and $[\gamma] \in \mathcal{V}_pM$ that:

$$\mathrm{D}_\gamma(f) = (f \circ \gamma)'(0) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (\varphi \circ \gamma)'(0)$$

Now suppose $\gamma_1, \ldots, \gamma_{n+1}$ are all in $\mathcal{C}(M,p)$. Then for any $c_1, \ldots, c_{n+1}$ we have that:

$$\left(\sum_{i=1}^{n+1} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \left(\sum_{i=1}^{n+1} c_i(\varphi \circ \gamma_i)'(0)\right)$$

Yet because $\mathbb{R}^n$ is an n-dimensional vector space, we know there exists $c_1, \ldots, c_{n+1}$ which aren't all zero such that $\sum_{i=1}^{n+1} c_i(\varphi \circ \gamma_i)'(0) = 0 \in \mathbb{R}^n$.

And then we'll have $\left(\sum_{i=1}^{n+1} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot 0 = 0$ for all $f \in C^1(M, \mathbb{R})$. This proves that $\dim(T_pM) \leq n$.

As for the other inequality, note that because $\varphi(U)$ is open in $(\mathbb{R}_{\geq 0})^n$, we can guarentee if $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$ then there exists $R > 0$ with $\varphi(p) + te_i \in \varphi(U)$ for all $t \in [0, R]$ and each i. So, I define $\gamma_i : [0, R] \to M$ by $\gamma(t) := \varphi^{-1}(\varphi(p) + te_i)$. Then for any $c_1, \ldots, c_n$ we have that:

$$\left(\sum_{i=1}^{n} c_i D_{\gamma_i}\right)(f) = \nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot (c_1, \ldots, c_n)$$

At last, we just need to show that if $\vec{c} := (c_1, \ldots, c_n) \neq 0$ then there exists some $f \in C^1(M, \mathbb{R})$ with $\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \vec{c} \neq 0$. Fortunately, we know from the proof of the lemma on *page 1047* how to go a step further and find $f \in C^r(M, \mathbb{R})$ with $\nabla(f \circ \varphi^{-1})(\varphi(p)) \cdot \vec{c} \neq 0$. Therefore:

- $D_{\gamma_1}, \ldots, D_{\gamma_n}$ is a collection of linearly independent vectors in $S \subseteq \mathcal{D}_pM$.
- If we restrict the domains of all the derivations in T_pM to just $C^r(M, \mathbb{R})$ then we still have that $D_{\gamma_1}, \ldots, D_{\gamma_n}$ are linearly independent vectors.

(b) Let $F : M \to N$ be a C^1 map. Then let $S' := \{D_\gamma : [\gamma] \in \mathcal{V}_{F(p)}N\}$ and let $T_{F(p)}N$ be the subspace of $\mathcal{D}_{F(p)}N$ spanned by S'.

[Note that if $F(p)$ is not a corner point of N, then this definition agress with our definition from *page 1051* because $T_{F(p)}N$ would equal S'].

Now define $\mathrm{d}F_p : \mathcal{D}_pM \to \mathcal{D}_{F(p)}N$ by $\mathrm{d}F_p(v)f := v(f \circ F)$. Then we have that $\mathrm{d}F_p(T_pM) \subseteq T_{F(p)}N$.

Proof:

It suffices to show that $\mathrm{d}F_p(S) \subseteq S'$. Fortunately, given any $D_\gamma \in S$ we have that $\mathrm{d}F_p(D_\gamma)f = D_\gamma(f \circ F) = (f \circ F \circ \gamma)'(0) = D_{F \circ \gamma}f$. Therefore $\mathrm{d}F_p(D_\gamma) = D_{F \circ \gamma} \in S'$. ■

Based on the prior lemma, we now have a way of defining T_pM when p is a corner point. Also by part (b), we know that the differential of $F : M \to N$ at p gives a well defined linear map $T_pM \to T_{F(p)}N$.

The proofs for propositions 3.6, 3.8, and 3.9 on pages *1031* and *1033-1034* can now be repeated verbatim for our modified definition of T_pM for when p is a corner chart and M is a C^r manifold with $r \neq \infty$ necessarily. Similarly, nothing on *pages 1051-1053* breaks down with our definition for this special case either. Finally, proposition 4.1 on *pages 1054-1056* as exercise 4-2 on the prior page still work by identical reasoning.

And with that I'll now allow my manifolds to have corners again by default.

Next, I want to talk about product manifolds. Let $M_1, \ldots, M_k$ be C^r-manifolds of dimensions $n_1, \ldots, n_k$ respectively. Then let $M := M_1 \times \cdots \times M_k$ be their product manifold, and for each i let $\pi_i : M \to M_i$ be the projection map $(p_1, \ldots, p_k) \mapsto p_i$.

Remark: Given any finite collection of C^r manifolds $M_1, \ldots, M_k$ with corners, the standard product C^r-structure on $M := M_1 \times \cdots \times M_k$ defined on *page 1038* is still well-defined. Furthermore, the proof of the proposition on *page 1039* still works verbatim. Thus the category of C^r manifolds with corners (which I'll denote **Diff**$_c^r$) does admit products over any finite collection of objects. In particular, we can see that $(p_1, \ldots, p_k)$ is a corner point of M if at least one p_i is a corner point of M_i or there are two components p_i and p_j which are boundary points of M_i and M_j respectively.

Proposition 3.14: Let $p = (p_1, \ldots, p_k) \in M$. Then the map $\alpha : T_pM \to \bigoplus_{i=1}^k T_{p_i}M_i$ defined by $\alpha(v) := (\mathrm{d}(\pi_1)_p(v), \ldots, \mathrm{d}(\pi_k)_p(v))$ is an isomorphism.

Proof:

For each i let us define $\iota_i : M_i \to M$ by $\iota_i(q) := (p_1, \ldots, p_{i-1}, q, p_{i+1}, \ldots, p_k)$. Then, we define $\beta : \bigoplus_{i=1}^k T_{p_i}M_i \to T_pM$ by $\beta(v_1, \ldots, v_k) := \sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)$. I claim that $\beta = \alpha^{-1}$. To see why, first note that:

$$\begin{aligned}
(\alpha \circ \beta)(v_1, \ldots, v_k) &= \alpha\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right) \\
&= \left(\mathrm{d}(\pi_1)_p\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right), \ldots, \mathrm{d}(\pi_k)_p\left(\sum_{i=1}^k \mathrm{d}(\iota_i)_{p_i}(v_i)\right)\right) \\
&= \left(\sum_{i=1}^k \mathrm{d}(\pi_1)_p(\mathrm{d}(\iota_i)_{p_i}(v_i)), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k)_p(\mathrm{d}(\iota_i)_{p_i}(v_i))\right) \\
&= \left(\sum_{i=1}^k \mathrm{d}(\pi_1 \circ \iota_i)_{p_i}(v_i), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k \circ \iota_i)_{p_i}(v_i)\right)
\end{aligned}$$

But now $\pi_j \circ \iota_i$ is a constant map if $j \neq i$. As a result, if $v \in T_{p_i}M_i$ and $f \in C^1(M_j, \mathbb{R})$ then $f \circ \pi_j \circ \iota_i$ is a constant function from M_i to $\mathbb{R}$ and in turn:

$$\mathrm{d}(\pi_j \circ \iota_i)_{p_i}(v_i)f = v_i(f \circ \pi_j \circ \iota_i) = 0.$$

Since f was arbitrary, this shows that if $j \neq i$ then $\mathrm{d}(\pi_j \circ \iota_i)_{p_i}(v)$ is the zero map. And since v was arbitary, we can conclude $\mathrm{d}(\pi_j \circ \iota_i)_{p_i}$ is also the zero map. Meanwhile, $\pi_i \circ \iota_i = \mathrm{Id}_{M_i}$. Hence, we can conclude that:

$$\begin{aligned}
\left(\sum_{i=1}^k \mathrm{d}(\pi_1 \circ \iota_i)_{p_i}(v_i), \ldots, \sum_{i=1}^k \mathrm{d}(\pi_k \circ \iota_i)_{p_i}(v_i)\right) &= (\mathrm{d}(\mathrm{Id}_{M_1})_{p_1}(v_1), \ldots, \mathrm{d}(\mathrm{Id}_{M_k})_{p_k}(v_k)) \\
&= (v_1, \ldots, v_k).
\end{aligned}$$

With that, we've proven that β is a right inverse of α. Based on the fact that α has a right inverse, we know that α is a surjective linear map. And since T_pM and $\bigoplus_{i=1}^k T_{p_i}M_i$ are both $(n_1 + \ldots + n_k)$-dimensional vector spaces and α is a linear map, we can conclude more strongly that α must be a bijection. And finally, any right inverse of a bijective function must also be a left inverse. Therefore, $\beta = \alpha^{-1}$. ■

The map α in the prior proposition is taken to define a standard identification between T_pM and $\bigoplus_{i=1}^k T_{p_i}M_i$.

7/24/2026

Proposition 4.6 (Properties of local C^k-diffeomorphisms):

(a) Let M, N, P be C^r manifolds and suppose $k \leq r$. If $F : M \to N$ and $G : N \to P$ are local C^k-diffeomorphisms, then so is $G \circ F : M \to P$.

Why?
Consider any $p \in M$. Since F is a local C^k-diffeomorphism, we know there exists an open set $U \subseteq M$ containing p and an open set $V \subseteq N$ such that $F|_U$ is a C^k-diffeomorphism from U to V. Next, since G is a local C^k-diffeomorphism, we know that there is an open set $V' \subseteq N$ containing $F(p)$ and an open set $W' \subseteq P$ such that $G|_{V'}$ is a C^k-diffeomorphism from V' to W'. And now we set $V_0 := V \cap V'$, $W_0 := G(V_0)$, and $U_0 := F^{-1}(V_0) \cap U$. We know that U_0 and W_0 are open sets in M and P respectively, that $p \in U_0$, and that $(G \circ F)$ is a C^k diffeomorphism from U_0 to W_0.

(b) Suppose $M_1, \ldots, M_k$ are C^r manifolds with $M := M_1 \times \cdots \times M_k$, $N_1, \ldots, N_k$ are also C^r manifolds with $N := N_1 \times \cdots \times N_k$, and $F_i : M_i \to N_i$ is a local C^s diffeomorphism for all $1 \leq i \leq k$ (where $s \leq r$). Then $F_1 \times \cdots \times F_k : M \to N$ is a local C^s diffeomorphism.

Why?
Let $(p_1, \ldots, p_k)$ be any point in M. For each i we can pick open sets $U_i \subseteq M_i$ containing p and $V_i \subseteq N_i$ such that $F|_{U_i} : U_i \to V_i$ is a C^s diffeomorphism. And now $U := U_1 \times \cdots \times U_k$ is an open subset of M containing $(p_1, \ldots, p_k)$ and $V := V_1 \times \cdots V_k$ is an open subset of N. I claim $F := F_1 \times \cdots \times F_k$ is a C^s diffeomorphism from U to V. This is because of the following lemma.

Lemma: Suppose $M_1, \ldots, M_k$ are C^r manifolds with $M := M_1 \times \cdots \times M_k$, $N_1, \ldots, N_k$ are also C^r manifolds with $N := N_1 \times \cdots \times N_k$, and $F_i : M_i \to N_i$ is a C^s diffeomorphism for all $1 \leq i \leq k$. Then $F_1 \times \cdots \times F_k : M \to N$ is a C^s diffeomorphism.

Proof:
It's incredibly straightforward to show that $F_1 \times \cdots \times F_k$ is a C^s map. Also, for each i let G_i be the inverse map of F_i. Then as each G_i is C^s, we can conclude by identical reasoning that $G_1 \times \cdots \times G_k : N \to M$ is a C^s map. And also it is easy to check that

$$G_1 \times \cdots \times G_k = (F_1 \times \cdots \times F_k)^{-1}.$$

(c) Let M and N be C^r manifolds, and suppose $F : M \to N$ is a local C^k diffeomorphism (where $k \leq r$). Then F is a local homeomorphism and an open map.

Proof:
Since diffeomorphisms are homeomorphisms, the first claim should be obvious. Next suppose $U \subseteq M$ is any open set. Then for any $p \in U$ we need to prove that $F(p)$ is in the interior of $F(U)$. Fortunately, as F is a local diffeomorphism we know there exists an open set $U_p \subseteq M$ containing p and an open set $V_p \subseteq N$ such that $F|_{U_p} : U_p \to V_p$

is a diffeomorphism. Next, as $U \cap U_p$ is an open subset of U_p, we know $F|_{U_p}(U \cap U_p)$ must be an open subset of V_p. So, $V := V_p \cap F(U \cap U_p)$ is an open set in N which is contained in $F(U)$. Also, $F(p) \in V$. ■

I'm bored and don't want to prove anymore parts of the prior proposition. So I'm moving on.

Proposition 4.8: Suppose M is C^r manifold without a boundary, N is any other C^r manifold, and F is a C^k function (where $k \leq r$).

(a) F is a local C^k-diffeomorphism if and only if it is both an immersion and submersion.

$(\Longrightarrow)$
Given any $p \in M$, there is an open set $U \subseteq M$ containing p such that $F(U)$ is an open set and F is a C^k-diffeomorphism from U to $F(U)$. But then in turn we must have that $\mathrm{d}(F|_U)_p : T_pU \to T_{F(p)}(F(U))$ is an invertible map. Equivalently, this means that $\mathrm{d}F_p : T_pM \to T_pN$ is invertible. And since was p was arbitrary, we can conclude that $\mathrm{rank}(F) = \dim(M) = \dim(N)$. This shows that F is an immersion and a submersion.

$(\Longleftarrow)$
Conversely, suppose F is an immersion and a submersion. Then we must have that $\mathrm{d}F_p$ is invertible for all $p \in M$. By the inverse function theorem for manifolds as well as exercise 4-2 on *pages 1056-1057*, we thus know that $F(M) \subseteq \mathrm{Int}(N)$ and F is a local C^k-diffeomorphism.

(b) If $\dim(M) = \dim(N)$ and F is either an immersion or submersion, then F is a local C^k-diffeomorphism.

Why?
If $\dim(M) = \dim(N)$ then F is an immersion if and only if F is a submersion. ■

Exercise 4.10: Suppose M, N, and P are C^r manifolds and $F : M \to N$ is a local C^k diffeomorphism.

(a) If $G : P \to M$ is a continuous map then G is C^k if and only if $F \circ G$ is C^k.

$(\Longrightarrow)$
This is obvious.

$(\Longleftarrow)$
Let $p \in P$. Since $F \circ G$ is C^k, we know that there are smooth charts (U, φ) on P containing p and (V, ψ) on N with $(F \circ G)(U) \subseteq V$ and $\psi \circ F \circ G \circ \varphi^{-1}$ a C^k map. Next, because F is a local C^k diffeomorphism we know that there are open sets $W' \subseteq M$ containing $G(p)$ and $V' \subseteq N$ such that F is a C^k diffeomorphism from W' to V'. By shrinking W' and V', we can assume without loss of generality that $V' \subseteq V$ and that there is a smooth chart (W', η) on M. Finally, because G is continuous, we know that $U' := G^{-1}(W')$ is an open set of P. So, we set $U_0 := U \cap U'$.

Let $H = (F|_{W'})^{-1}$. Then (U_0, φ) is a smooth chart on P containing p, (W', η) is a smooth chart on M with $F(U_0) \subseteq W'$, and:

$$\eta \circ G \circ \varphi^{-1} = (\eta \circ H \circ \psi^{-1}) \circ (\psi \circ F \circ G \circ \varphi^{-1}) \text{ is a } C^k \text{ map.}$$

(b) If F is surjective and $G : N \to P$ is any map, then G is C^k if and only if $G \circ F$ is C^k.

$(\Longrightarrow)$
Again this is obvious.

$(\Longleftarrow)$
Consider any $p \in N$. Since F is surjective, we know there exists $q \in M$ such that $F(q) = p$. Next, since $G \circ F$ is a C^k map, we know there is a smooth chart (U, φ) on M containing q and a smooth chart (V, ψ) on P such that $(G \circ F)(U) \subseteq V$ and $\psi \circ G \circ F \circ \varphi^{-1}$ is a C^k map. But now as F is a local C^k diffomorphism, we can find an open set $U' \subseteq M$ containing q and an open set $W' \subseteq N$ such that F is a C^k diffeomorphism from U' to W'. Finally, by shrinking U' and W', we can assume that $U' \subseteq U$ and that there is a smooth chart (W', η) on N.

But now W' is an open set containing $F(p)$ such that $W' = F(U') \subseteq F(U) \subseteq V$. Also let $H = (F|_{U'})^{-1}$. Then:

$$\psi \circ G \circ \eta^{-1} = (\psi \circ G \circ F \circ \varphi^{-1}) \circ (\varphi \circ H \circ \eta^{-1}) \text{ is a } C^k \text{ map. } \blacksquare$$

7/26/2026

Welp, I've been fearing this day for a long time but I finally need to actually understand the rank theorem.

Theorem 4.12 (The Rank Theorem): Suppose M and N are C^k manifolds *without boundaries* of dimension m and n respectively, and also suppose $F : M \to N$ is a C^k map of constant rank r. Then for each $p \in M$ there exists a smooth chart (U, φ) on M with $p \in U$ as well as a smooth chart (V, ψ) on N with $F(U) \subseteq V$ such that $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ has the form $\widehat{F}(x_1, \ldots, x_r, x_{r+1}, \ldots, x_m) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{n-r \text{ many zeros}})$.

As a side note, if $k_1 > k_2$ in $\mathbb{Z}_{\geq 1} \cup \{\infty\}$ then every C^{k_1} structure on a manifold M is contained in a unique C^{k_2} structure. Therefore, it's not too too restrictive to assume F is as smooth a map as the structure on the manifold is smooth. That said, if I didn't make this restriction clear than one might falsely think that this theorem proves every constant rank C^1 map between C^k manifolds without boundaries is actually a C^k map.

Proof:
(Step 0: Simplying our notation)
Firstly, by choosing local coordinates around p it suffices to assume M is an open set $U \subseteq \mathbb{R}^m$ and N is an open set $V \subseteq \mathbb{R}^m$.

To see why, consider picking smooth charts $(\widetilde{U}, \widetilde{\varphi})$ on M containing p and $(\widetilde{V}, \widetilde{\psi})$ on N such that $F(\widetilde{U}) \subseteq \widetilde{V}$. Then set $U := \widetilde{\varphi}(\widetilde{U})$ and $V := \widetilde{\psi}(\widetilde{V})$ and note that U and V are open in $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively. Once we've shown that the theorem holds for $\widetilde{\psi} \circ F \circ (\widetilde{\varphi})^{-1} : U \to V$, then we can just find two C^k diffeomorphisms $\varphi : U_0 \to \varphi(U_0)$ and $\psi : V_0 \to \psi(V_0)$ such that $\widetilde{\varphi}(p) \subseteq U_0 \subseteq U$, $V_0 \subseteq V$, and $\widetilde{\psi} \circ F \circ (\widetilde{\varphi})^{-1}(U_0) \subseteq V_0$, and $(\psi \circ \widetilde{\psi}) \circ F \circ ((\widetilde{\varphi})^{-1} \circ \varphi^{-1})$ has the desired form. And since $(\widetilde{\varphi}^{-1}(U_0), \varphi \circ \widetilde{\varphi})$ is a smooth chart on M and $(\widetilde{\psi}^{-1}(V_0), \psi \circ \widetilde{\psi})$ is a smooth chart on N, we'll have shown our theorem holds at p for the map F.

Next, note that since $\mathrm{D}F(p)$ has rank r, we know the Jacobian matrix has some $r \times r$ invertible submatrix. By reordering our coordinates, we can assume without loss of generality that the $r \times r$ invertible submatrix makes up the top left portion of $\mathrm{D}F(p)$. So, let us rename the standard coordinates on $\mathbb{R}^m$ and $\mathbb{R}^n$ as:

$(x, y) := (x_1, \ldots, x_r, y_{r+1}, \ldots, y_m)$ and $(v, w) := (v_1, \ldots, v_r, w_{r+1}, \ldots, w_n)$.

By applying a translation to the open sets U and V, we can without loss of generality assume $p = (0,0)$ and $F(p) = (0,0)$. Also, we write $F(x,y) := (Q(x,y), R(x,y))$ where $Q : U \to \mathbb{R}^r$ and $R : U \to \mathbb{R}^{n-r}$ are C^k functions. Therefore:

$$\mathrm{D}F(0,0) = \begin{bmatrix} \left[\frac{\partial Q_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}} & \left[\frac{\partial Q_i}{\partial y_j}(0,0)\right]_{\substack{1 \le i \le r \\ r+1 \le j \le m}} \\ \left[\frac{\partial R_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le n-r \\ 1 \le j \le r}} & \left[\frac{\partial R_i}{\partial y_j}(0,0)\right]_{\substack{1 \le i \le n-r \\ r+1 \le j \le m}} \end{bmatrix}$$

where $\left[\frac{\partial Q_i}{\partial x_j}(0,0)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}}$ is an invertible matrix.

<u>(Step 1: Getting φ)</u>
Now define $\varphi : U \to \mathbb{R}^m$ by $\varphi(x,y) = (Q(x,y), y)$. This will be C^k map whose total derivative is given by:

$$\mathrm{D}\varphi(x,y) = \begin{bmatrix} \left[\frac{\partial Q_i}{\partial x_j}(x,y)\right]_{\substack{1 \le i \le r \\ 1 \le j \le r}} & \left[\frac{\partial Q_i}{\partial y_j}(x,y)\right]_{\substack{1 \le i \le r \\ r+1 \le j \le m}} \\ \mathbf{0} & \mathbf{I}_{m-r} \end{bmatrix}$$

where $\mathbf{I}_{m-r}$ is the $(m-r) \times (m-r)$ identity matrix. Importantly, we can specifically see that all the columns of $\mathrm{D}\varphi(0,0)$ are linearly independent. Hence, $\mathrm{D}\varphi(0,0)$ is invertible. And by the inverse function theorem we know there are connected open sets $U_0 \subseteq U$ and $\widehat{U}_0 \subseteq \mathbb{R}^m$ satisfying that $(0,0) \in U_0$, $\varphi(0,0) = (0,0) \in \widehat{U}_0$, and $\varphi : U_0 \to \widehat{U}_0$ is a C^k-diffeomorphism. By shrinking U_0 and $\widehat{U}_0$ we can without loss of generality assume that $\widehat{U}_0 = (-\varepsilon, \varepsilon)^m$ for some $\varepsilon > 0$.

Now we write the inverse map $\varphi^{-1} : \widehat{U}_0 \to U_0$ as $\varphi^{-1}(x,y) = (A(x,y), B(x,y))$ where $A : \widehat{U}_0 \to \mathbb{R}^r$ and $B : \widehat{U}_0 \to \mathbb{R}^{m-r}$ are C^k maps. Importantly, we know that:

$$(x,y) = \varphi(A(x,y), B(x,y)) = (Q(A(x,y), B(x,y)),\ B(x,y))$$

By comparing y-components we can see that $B(x,y) = y$. Hence, φ^{-1} has the form

$\varphi^{-1}(x, y) = (A(x, y),\ y)$. Meanwhile, by comparing x-components we can see that $x = Q(A(x, y),\ y)$. Therefore, upon letting $\widetilde{R}(x, y) := R(A(x, y), y)$ then we have that:

$$(F \circ \varphi^{-1})(x, y) = (x,\ \widetilde{R}(x, y))$$

<u>(Step 2: Simplifying $\widetilde{R}$)</u>
Consider that the Jacobian matrix of $F \circ \varphi^{-1}$ at any $(x, y) \in \widehat{U}_0 = (-\varepsilon, \varepsilon)^m$ is given by:

$$\mathrm{D}(F \circ \varphi^{-1})(x, y) = \begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \left[\frac{\partial \widetilde{R}_i}{\partial x_j}(x, y)\right]_{\substack{1 \leq i \leq n-r \\ 1 \leq j \leq r}} & \left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x, y)\right]_{\substack{1 \leq i \leq n-r \\ r+1 \leq j \leq m}} \end{bmatrix}$$

Yet we also have that $\mathrm{D}(F \circ \varphi^{-1})(x, y) = \mathrm{D}F(\varphi^{-1}(x, y))\mathrm{D}(\varphi^{-1})(x, y)$. By assumption, we know that $\mathrm{rank}(\mathrm{D}F(\varphi^{-1}(x, y))) = r$ for all $(x, y) \in \widehat{U}_0$. Meanwhile, since φ^{-1} is a diffeomorphism, we know that $\mathrm{D}(\varphi^{-1})(x, y)$ defines a bijective linear map for all $(x, y) \in \widehat{U}_0$. It follows that $\mathrm{rank}(\mathrm{D}(F \circ \varphi^{-1})(x, y))$ will still equal r for all $(x, y) \in \widehat{U}_0$.

Yet the only way this matrix:

$$\begin{bmatrix} \mathbf{I}_r & \mathbf{0} \\ \left[\frac{\partial \widetilde{R}_i}{\partial x_j}(x, y)\right]_{\substack{1 \leq i \leq n-r \\ 1 \leq j \leq r}} & \left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x, y)\right]_{\substack{1 \leq i \leq n-r \\ r+1 \leq j \leq m}} \end{bmatrix},$$

can ever have rank r is if $\left[\frac{\partial \widetilde{R}_i}{\partial y_j}(x, y)\right]_{\substack{1 \leq i \leq n-r \\ r+1 \leq j \leq m}} = \mathbf{0}$.

> This is because the first r columns are already linearly independent, and the intersection of their span with the span of the last $m - r$ columns is just the trivial subspace $\{0\}$.

Consequently, we know that $\widetilde{R}$ must be constant with respect to y. In other words, there exists some C^k map $S : (-\varepsilon, \varepsilon)^r \to \mathbb{R}^{n-r}$ such that $\widetilde{R}(x, y) = S(x)$. And with that, we can write:

$$(F \circ \varphi^{-1})(x, y) = (x, S(x))$$

<u>(Step 3: Getting ψ)</u>
Let $V_0 := \{(v, w) \in V : (v, 0) \in \widehat{U}_0\}$. Then V_0 will be an open subset of V containing $F(0, 0) = (0, 0)$. Also because $\widehat{U}_0$ is a cube and $(F \circ \varphi^{-1})(x, y) = (x, S(x))$, we can see that $(F \circ \varphi^{-1})(\widehat{U}_0) \subseteq V_0$. (And this also implies $F(U_0) \subseteq V_0$).

To finish off, define $\psi : V_0 \to \mathbb{R}^n$ by $\psi(v, w) = (v, w - S(v))$. Then ψ is a C^k map which is also a diffeomorphism onto its image. After all, by the invariance of domain theorem we know its image is an open subset of $\mathbb{R}^n$. Also, its inverse map is precisely given by $\psi(v, w) = (v, w + S(v))$.

And now just note that $(\psi \circ F \circ \varphi^{-1})(x, y) = \psi(x, S(x)) = (x, S(x) - S(x)) = (x, 0)$.
■

<u>Corollary 4.13:</u> Let M and N be C^k manifolds without boundaries and let $F : M \to N$ be a C^k map. Also suppose M is connected. The following are equivalent.

(a) For each $p \in M$ there exists smooth charts (U, φ) on M and (V, ψ) on N with $p \in U$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi$ a linear map.

(b) F has constant rank.

Proof:
Firstly suppose (a) holds and for every $r \leq N := \min(\dim(M), \dim(N))$ let $A_r \subseteq M$ be the set of all $p \in M$ at which F has rank r. Trivially each A_r is disjoint with $\bigcup_{r=0}^{N} A_r = M$. By (a) we also know each A_r is an open set. By the connectedness of M we can thus conclude there exists a single r with $A_r = M$. So, F has constant rank on M.

Conversely, if (b) is true than the rank theorem tells us how to find the smooth charts described in (a). ■

<u>Theorem 4.14:</u> Let M, N be C^k manifolds without boundaries and let $F : M \to N$ be a C^1 map of constant rank.

(a) If F is surjective then F is a submersion.

Proof:
Let $m = \dim(M)$ and $n = \dim(N)$, and suppose F has constant rank r. We'll suppose to the contrary that F is *not* a submersion. Thus $r < n$.

Now without loss of generality we can assume $k = 1$ (i.e. that M and N are C^1 manifolds).

This is because the C^k structures $\mathcal{A}_M$ and $\mathcal{A}_N$ that M and N are equipped with are both contained in unique C^1 structures $\mathcal{B}_M$ and $\mathcal{B}_N$ respectively. Furthemore, a map $f : M \to \mathbb{R}$ is C^1 with respect to $\mathcal{A}_M$ if and only if it is C^1 with respect to $\mathcal{B}_M$. And a similar story holds for any $f : N \to \mathbb{R}$. It follows that the tangents spaces to M and N stay identical when each is equipped with either structure. In turn $F : (M, \mathcal{A}_M) \to (N, \mathcal{A}_N)$ is a submersion if and only if $F : (M, \mathcal{B}_M) \to (N, \mathcal{B}_N)$ is a submersion.

Due to the prior reasoning, the rank theorem now applies. So, for all $p \in M$ there exists a smooth chart (U_p, φ_p) on M containing p and a smooth chart (V_p, ψ_p) on N with $F(U_p) \subseteq V_p$ and $\widehat{F}_p : \psi \circ F \circ \varphi^{-1}$ having the form:

$$\widehat{F}_p(x_1, \ldots, x_r, x_{r+1}, \ldots, x_m) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{\text{nonzero \# of zeros}}).$$

Since M is an LCH space, we can for each p pick a precompact open set $W_p \subseteq M$ such that $p \in W_p \subseteq \overline{W_p} \subseteq U_p$. Then as $\overline{W_p}$ is compact and N is Hausdorff, we know that $F(\overline{W_p})$ is a closed subset of N containing $F(W_p)$. Moreover, $F(\overline{W_p})$ has an empty interior. After all, let $\widehat{U}_p := \varphi_p(U_p)$ and $\widehat{V}_p := \psi_p(V_p)$. Then we have that $(\psi \circ F)(\overline{W_p}) \subseteq \widehat{F}_p(\widehat{U}_p) \subseteq$ (a hyperplane) $\cap\, \widehat{V}_p$. Since the latter set has an empty interior relative to $\widehat{V}_p$, we can conclude that $(\psi \circ F)(\overline{W_p})$ has an empty interior relative to $\widehat{V}_p$ as well. Then since ψ is a homeomorphism from V_p to $\widehat{V}_p$, we can conclude that $F(\overline{W_p})$ has an empty interior relative to V_p. And since $F(\overline{W_p})$ is a closed subset of N containing $F(W_p)$ and contained in V_p, we can finally conclude that $F(W_p)$ is nowhere dense.

But now to get our contradiction, note that because M is second countable there must be a sequence $\{p_\ell\}_{\ell\in\mathbb{N}}$ such that $M = \bigcup_{\ell\in\mathbb{N}} W_{p_\ell}$. Since F is surjective, we in turn know that $N = F(M) = \bigcup_{\ell\in\mathbb{N}} F(W_{p_\ell})$. So, we've proven that N is a countable union of nowhere dense sets. In other words N is meager. Yet N is an LCH space and so this contradicts the Baire category theorem. [See *pages 540-541*.]

(b) If F is injective then F is an immersion.

Proof:
Let m, n, and r be as in the prior proof, and this time suppose to the contrary that F is *not* an immersion. For identical reasons as in the last proof, we can assume without loss of generality that $k = 1$ (i.e. that M and N are C^1 manifolds). And this is important because it lets us apply the rank theorem.

Consider any $p \in M$ and let (U, φ) and (V, ψ) be smooth charts on M such that $p \in U$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ has the form:

$$\widehat{F}(x_1, \ldots, x_r, \underbrace{x_{r+1}, \ldots, x_m}_{\text{nonzero } \# \text{ of coordinates}}) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{n-r \text{ many zeros}}).$$

But now F being injective implies that $\widehat{F}$ is also injective. Simultaneously, write $\varphi(p) = (p_1, \ldots, p_r, p_{r+1}, \ldots, p_m)$. Since $\varphi(U)$ is open in $\mathbb{R}^n$ we know for some small enough $\varepsilon > 0$ that $\widehat{q} := (p_1, \ldots, p_r, p_{r+1} + \varepsilon, \ldots, p_m) \in \varphi(U)$. But now if we let $q = \varphi^{-1}(q)$ we have that $(\psi \circ F)(p) = \widehat{F}(\varphi(p)) = \widehat{F}(\varphi(q)) = (\psi \circ F)(q)$. Since ψ is injective, we in turn have that $F(p) = F(q)$. Yet $p \neq q$ since φ^{-1} is injective. So, this all contradicts that F is injective. ■

Corollary: Let M, N be C^r manifolds without boundaries and let $F : M \to N$ be a C^k map of constant rank (where $k \leq r$). If F is a bijection then F is a C^k diffeomorphism.

Proof:
By the prior theorem, F being a bijection means that F is both a submersion and an immersion. It then follows by proposition 4.8 on *page 1061* that F is a local C^k diffeomorphism. Yet finally, we can easily see by the pasting lemma that any local C^k diffeomorphism that is also bijective is a (global) C^k diffeomorphism. So, F is a C^k diffeomorphism. ■

7/27/2026

I mentioned a few days ago that my construction of the tangent space to a corner point might not be optimal. After talking to a colleague, I can now say that while nothing I've written is wrong, there is a cleaner equivalent way of defining the tangent space. This improved way will also be the way that generalizes to analytic and complex manifolds. So, it wouldn't hurt to go over now.

Firstly, I'm going to fix an oversight I priorly made on *page 1024*. I already deleted the false statement on that page but I did leave a note explaining where my deleted remark went wrong.

Firstly, since it will make my life easier and serve as a good review, I want to generalize the universal property of tensor products of module. Note that this will only use knowledge from math 200B.

Given a collection of A-modules $M_1, \ldots, M_k$, we say their tensor product $\bigotimes_{i=1}^k M_i$ is an A-module equipped with a k-linear map $\iota : \prod_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i$ such that for any k-linear map $T : \prod_{i=1}^k M_i \to N$ the following diagram holds:

$$\begin{tikzcd}
 & & & m_1\otimes\cdots\otimes m_k := \iota(m_1,\ldots,m_k) \arrow[dd, mapsto] \\
 & & \bigotimes_{i=1}^k M_i \arrow[d, dashed, "\exists! f_T"] & \\
(m_1,\ldots,m_k) \arrow[uurrr, mapsto, bend left] \arrow[rrr, mapsto, bend right] & \prod_{i=1}^k M_i \arrow[ur, "\iota"] \arrow[r, "T"'] & N & T(m_1,\ldots,m_k)
\end{tikzcd}$$

In math 200b I learned how to construct tensor products of 2 modules and I also showed that $M_1 \otimes (M_2 \otimes M_3) \cong (M_1 \otimes M_2) \otimes M_3$. But for the sake of better notation and easier thinking, I want to be able to talk about tensor products of any finite number of modules. Fortunately, by standard reasoning we can see that any two A-modules satisfying the universal property in the above diagram are isomorphic.

<u>Lemma A:</u> Let $k \geq 2$ and suppose $M_1, \ldots, M_k$ are all A-modules. Then there exists a module $\bigotimes_{i=1}^k M_i$ and a k-linear map $\iota : \prod_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i$ fulfilling the above universal property. Furthermore, $\iota(\prod_{i=1}^k M_i)$ is a generating / spanning set for $\bigotimes_{i=1}^k M_i$.

Proof:

Our base case of $k = 2$ is already covered. So let us now proceed by induction and assume that we've already defined $\bigotimes_{i=2}^k M_i$ and $\bar{\iota} : \prod_{i=2}^k M_i \to \bigotimes_{i=2}^k M_i$. Then let's define $M \coloneqq M_1 \otimes (\bigotimes_{i=2}^k M_i)$ and $\iota : \prod_{i=1}^k M_i \to M$ by:

$$\iota(m_1, \ldots, m_k) \coloneqq m_1 \otimes \bar{\iota}(m_2, \ldots, m_k).$$

It's easy to see that ι is a k-linear map. Furthemore, suppose $T : \prod_{i=1}^k M_i \to N$ is any k-linear map. Then for any fixed $m_1 \in M_1$, we can see that $T_{m_1} : \prod_{i=2}^k M_i \to N$ defined by $T_{m_1}(m_2, \ldots, m_k) \coloneqq T(m_1, \ldots, m_k)$ is a well-defined $(k-1)$-linear map. So by the universal property of $\bigotimes_{i=2}^k M_i$ and $\bar{\iota} : \prod_{i=2}^k M_i \to \bigotimes_{i=2}^k M_i$, we know that there exists a unique A-module homomorphism $f_{m_1} : \bigotimes_{i=2}^k M_i \to N$ such that $f_{m_1} \circ \bar{\iota} = T_{m_1}$.

Next, we define $S : M_1 \times (\bigotimes_{i=2}^k M_i) \to N$ by $S(m_1, t) := f_{m_1}(t)$. Since each f_{m_1} is A-linear, we know that S is A-linear in its second variable. Meanwhile, note that:

$$\begin{aligned} f_{am_1+a'm_1'}(\overline{\iota}(m_2, \dots, m_k)) &= T(am_1 + a'm_1', m_2, \dots, m_k) \\ &= aT(m_1, m_2, \dots, m_k) + a'T(m_1', m_2, \dots, m_k) \\ &= af_{m_1}(\overline{\iota}(m_2, \dots, m_k)) + a'f_{m_1'}(\overline{\iota}(m_2, \dots, m_k)). \end{aligned}$$

It follows S is A-linear in its first variable if its second variable is contained in $\overline{\iota}(\prod_{i=2}^k M_i)$. And since S is a A linear in its second variable and $\overline{\iota}(\prod_{i=2}^k M_i)$ spans all of $\bigotimes_{i=2}^k M_i$, we can in turn conclude that S is A-linear in its first variable regardless of the value of its second variable. And with that we've shown S is a bilinear map.

By the universal property of tensor products, we know there is a unique A-module homomorphism $f_T : M_1 \otimes (\bigotimes_{i=2}^k M_i) \to N$ such that the below diagram commutes:

$$\begin{tikzcd}
& & & m_1\otimes t \arrow[ddd, mapsto] \\
& & M_1 \otimes (\bigotimes_{i=2}^k M_i) \arrow[d, dashed, "\exists! f_T"] & \\
& M_1 \times (\bigotimes_{i=2}^k M_i) \arrow[ur] \arrow[r, "S"'] & N & \\
(m_1,t) \arrow[rrr, mapsto] \arrow[uuurrr, bend left, mapsto] & & & f_{m_1}(t)
\end{tikzcd}$$

I further claim that f_T is the unique A-module homomorphism making this diagram commute:

$$\begin{tikzcd}
& M \arrow[d, dashed, "f_T"] \\
\prod_{i=1}^k M_i \arrow[ur, "\iota"] \arrow[r, "T"'] & N
\end{tikzcd}$$

After all, the fact that this diagram commutes is simply because:

$$\begin{aligned} f_T(\iota(m_1, \dots, m_k)) &= f_T(m_1 \otimes \overline{\iota}(m_2, \dots, m_k)) \\ &= S(m_1, \overline{\iota}(m_2, \dots, m_k)) = f_{m_1}(\overline{\iota}(m_2, \dots, m_k)) \\ &= T(m_1, m_2, \dots, m_k). \end{aligned}$$

Meanwhile, if g was another A-module homomorphism making the diagram commute, we'd similarly be able to say that:

$$\begin{aligned} g(m_1 \otimes \overline{\iota}(m_2, \dots, m_k)) &= g((\iota(m_1, \dots, m_k))) \\ &= T(m_1, m_2, \dots, m_k) = f_{m_1}(\overline{\iota}(m_2, \dots, m_k)) \\ &= S(m_1, \overline{\iota}(m_2, \dots, m_k)). \end{aligned}$$

In turn since g is A-linear and $\overline{\iota}(\prod_{i=2}^k M_i)$ spans all of $\bigotimes_{i=2}^k M_i$, we can conclude that $g(m_1 \otimes t) = S(m_1, t)$ for all $(m_1, t) \in M_1 \times (\bigotimes_{i=2}^k M_i)$. By the uniqueness of f_T, we thus must have that $g = f_T$. And with that, we've proven that M equipped with ι satisfies the universal property of a tensor product of k many modules.

To finish off the proof, we already know our construction of $M_1 \otimes (\bigotimes_{i=2}^k M_i)$ in math 200b that that space is spanned by pure tensors of the form $m_1 \otimes t$ where $t \in \bigotimes_{i=2}^k M_i$. So, we can write any arbitrary element in M as a sum of pure tensors $\sum_{j=1}^{\ell} m_1^{(j)} \otimes t^{(j)}$.

(And via A-balance I'm letting $m_1^{(j)}$ absorb any scalars in this sum.)

But by induction we can then write $t^{(j)} = \sum_{j'=1}^{r_j} \overline{\iota}(m_2^{(j,j')}, \ldots, m_k^{(j,j')})$ (where again I'm letting the $m_i^{(j,j')}$ absorb any of the scalars). So:

$$\begin{aligned}\sum_{j=1}^{\ell} m_1^{(j)} \otimes t^{(j)} &= \sum_{j=1}^{\ell} \sum_{j'=1}^{r_j} m_1^{(j)} \otimes \overline{\iota}(m_2^{(j,j')}, \ldots, m_k^{(j,j')}) \\ &= \sum_{j=1}^{\ell} \sum_{j'=1}^{r_j} \iota(m_1^{(j)}, m_2^{(j,j')}, \ldots, m_k^{(j,j')}). \blacksquare\end{aligned}$$

As a side note: The prior proof can also be interpretted as saying that $M_1 \otimes (M_2 \otimes (\cdots \otimes (M_{k-1} \otimes M_k) \cdots))$ along with the k-linear map $\iota(m_1, \ldots, m_k) = m_1 \otimes (m_2 \otimes (\cdots \otimes (m_{k-1} \otimes m_k) \cdots))$ satisfies the universal property of a tensor product of k many A-modules.

Note that we can by identical reasoning then prove an analogous proposition as on *pages 726-728* if $M_1, \ldots, M_k$ are A-algebras (albeit with more cumbersome notation).

In other words, if $M_1, \ldots, M_k$ are A-algebras then $\bigotimes_{i=1}^k M_i$ is also an A-algebra when equipped with the multiplication operation:

$$(m_1 \otimes \cdots \otimes m_k)(m_1' \otimes \cdots \otimes m_k') = m_1 m_1' \otimes \ldots \otimes m_k m_k'$$

Similarly, given A-module homomorphisms $f_i : M_i \to M_i'$ we can show identically as on *page 700* that there is a unique A-module homomorphism $\bigotimes_{i=1}^k M_i \to \bigotimes_{i=1}^k M_i'$ such that $f(m_1 \otimes \cdots \otimes m_k) = f(m_1) \otimes \cdots \otimes f(m_k)$.

<u>Lemma B:</u> Suppose $N, M_1, \ldots, M_k$ are A-algebras and $f_i : M_i \to N$ is an A-algebra homomorphism for $1 \leq i \leq k$. Then there is a unique A-module homomorphism $f : \bigotimes_{i=1}^k M_i \to N$ such that $f(m_1 \otimes \cdots \otimes m_k) = \prod_{i=1}^k f_i(m_i)$.

Proof:

To start off, if we define $T : \prod_{i=1}^k M_i \to N$ by $T(m_1, \ldots, m_k) = \prod_{i=1}^k f_i(m_i)$ then we can easily see that T is k-linear. So by the universal property of $\bigotimes_{i=1}^k M_i$, we know there is a unique A-module homomorphism $f : \bigotimes_{i=1}^k M_i \to N$ such that:

$$f(m_1 \otimes \cdots \otimes m_k) = \prod_{i=1}^k f_i(m_i).$$

What we need to do now though is show that f also happens to be algebra homomorphism. This is not difficult but involves a lot of tedious symbol pushing which I don't want to type out. $\blacksquare$

Another observation to make is that if each $f_i : M_i \to N$ is unital then f is also unital.

7/31/2026

This tangent has actually become significantly longer and deeper than I originally thought it would be. I just don't know enough about algebras. So at this point I'm going to start following the textbook *Multilinear Algebra* by Douglas Northcott. See *citation 12* for more information. Also I'm going to switch back to the normal color scheme.

For the time being, I'll let R denote a fixed commutative unital ring. I'm letting R as opposed to A denote the ring both because the ring I will care about eventually is just $\mathbb{R}$, and also because this frees up A to refer to an R-algebra.

Let A be an R-algebra (which doesn't have to be commutative). We say a family $\{A_n\}_{n\in\mathbb{Z}}$ of R-submodules of A is a grading on A if $A = \bigoplus_{n\in\mathbb{Z}} A_n$ as an internal direct sum of R-modules and $a_r \in A_r$ and $a_s \in A_s$ implies that $a_r a_s \in A_{r+s}$.

- An R-algebra with a grading $\{A_n\}_{n\in\mathbb{Z}}$ is said to be a graded R-algebra. If all $A_n = \{0\}$ for $n < 0$ then we say our grading is nonnegative.
- Also any elements of A_n are called homogeneous of degree n.
- By definition, any $a \in A$ can be written uniquely as $\sum_{n\in\mathbb{Z}} a_n$ where each $a_n \in A$ and all but finitely many a_n are zero. We call the a_n in this sum the homogeneous coefficients of a.

Theorem 3.5: Let A be a unital graded R-algebra with the grading $\{A_n\}_{n\in\mathbb{N}}$. Then 1_A is homogeneous of degree 0.

Let us write $1_A = \sum_{n\in\mathbb{Z}} a_n$ as a sum of the homogeneous coefficients of 1_A. Then for any $m \in \mathbb{Z}$ we consider any $b_m \in A_m$. But now note that:

$$\sum_{n\in\mathbb{Z}} a_n b_m = 1_A b_m = b_m = b_m 1_A = \sum_{n\in\mathbb{Z}} b_m a_n$$

By comparing homogeneous coefficients, we get that $a_0 b_m = b_m = b_m a_0$ (and also that $a_n b_m = 0 = b_m a_n$ if $n \neq 0$). And thus since every $b \in A$ is a sum of $b_m \in A_m$ as m ranges over some finite subset of $\mathbb{Z}$, we can conclude for all $b \in A$ that $a_0 b = b = b a_0$. This shows that a_0 is an identity element of A. And thus $1_A = 1_A a_0 = a_0$. ■

Given a unital R-algebra A, I'll say $B \subseteq A$ is a subalgebra if it is also an R-algebra under the algebra operations of A and $1_B = 1_A$.

Corollary: Let A be a graded unital R-algebra with the grading $\{A_n\}_{n\in\mathbb{N}}$. Then A_0 is a subalgebra of A.

Why?

A_0 is an R-module and we know from the definition of a grading that if $a, b \in A_0$ then so is $ab \in A_0$. Finally, $1_A \in A_0$ lets us conclude A_0 is a subalgebra. ■

Let A be an R-algebra. Note that if $\{B_i\}_{i\in I}$ is a collection of subalgebras of A then so is $B := \bigcap_{i\in I} B_i$. As a result, given a subset S of A we let the R-subalgebra generated by S (denoted $\langle S\rangle$) to be the smallest subalgebra of A containing S. We can construct $\langle S\rangle$ as follows:

Let B be the collection of all finite linear combinations of finite products of elements of S (where a product of zero-many elements is interpretted as equalling 1_A). It is easy to justify that $B \subseteq \langle S \rangle$ and that B is itself a subalgebra of A which contains S. So, $\langle S \rangle \subseteq B$.

Lemma 3.1: Let $\{A_n\}_{n\in\mathbb{Z}}$ be a grading on a unital R-algebra A, and suppose that $A = \langle A_1 \rangle$. Then the grading on A is nonnegative and for all $n \geq 0$ we have that any $a \in A_n$ is a finite linear combinations of products of n many elements of A_1. In particular, this means that A_0 is generated as an R-module by 1_A. So, A_0 is contained in the center of A.

Proof:

A is generated as an R-module by all finite products of elements of A_1. Therefore, let $B_0 = R1_A$, let $B_n = \{0\}$ for $n < 0$, and finally for $n > 0$ let B_n be the R-module generated by all finite products of n many elements of A_1. We know $A = \sum_{n\in\mathbb{Z}} B_n$, and we can also easily see that $B_n \subseteq A_n$ for all n. The only way both statements can be true is if $A_n = B_n$ for all n. ■

Let A and B be unital graded R-algebras with gradings $\{A_n\}_{n\in\mathbb{N}}$ and $\{B_n\}_{n\in\mathbb{N}}$ respectively. We say $f : A \to B$ is a homomorphism of graded R-algebras if it is a unital R-algebra homomorphism and $f(A_n) \subseteq B_n$ for all n.

Lemma: If $f : A \to B$ is a homomorphism of graded R-algebras and a bijection, then $f(A_n) = B_n$ for all n.

Proof:

Consider any $n \in \mathbb{Z}$ and $b_n \in B_n$. Then write $f^{-1}(b_n) = \sum_{m\in\mathbb{Z}} a_m$ where $a_m \in A_m$ for each m. In turn, $b_n = f(f^{-1}(b_n)) = f(\sum_{m\in\mathbb{Z}} a_m) = \sum_{m\in\mathbb{Z}} f(a_m)$. So if $m \neq 0$ then we can compare homogeneous coefficients of b_n to get that $f(a_m) = 0$. Since f is injective, it follows that $a_m = 0$ for all $m \neq 0$. And so, we've shown that $f^{-1}(b_n) \in A_n$. In other words, $f^{-1}(B_n) \subseteq A_n$. And since $f(A_n) \subseteq B_n$, we can now conclude that $f(A_n) = B_n$. ■

Let $I \lhd A$ be a two-sided ideal of a graded unital R-algebra A (i.e. I is an R-submodule and a two-sided ideal of A). We say I is homogeneous if $a \in I$ implies that all the homogeneous components of A are also in I.

Recall the definition of sheaves, stalks, and germs given on *pages 1026-1027*. Then given any C^r manifold M and any $k \leq r$ in $\mathbb{Z}_{\geq 1} \cup \{\infty\}$, we consider the sheaf $\mathcal{F}$: **Op**$(M) \to \mathbb{R}$-**alg** such that $\mathcal{F}(U) = C^k(U)$ and for any open sets $V \subseteq U \subseteq M$ we have that $r_V^U(f) = f|_V$.

For any $p \in M$ we denote the stalk of $\mathcal{F}$ at p to be the set $C_p^k(M)$. Also, given any open set U and C^k map $f : U \to \mathbb{R}$, we let $[(U, f)]$ denote the germ of f in $C_p^k(M)$.

Lemma A: Suppose $[(U, f)]$ and $[(V, g)]$ are germs in $C_p^k(M)$. Then $[(U, f)] = [(V, g)]$ if and only if there is an open set $W \subseteq U \cap V$ such that $p \in W$ and $f|_W = g|_W$.

$(\Longleftarrow)$

By the definition of a direct limit, we know $[(U, f)] = [(W, f|_W)]$ and $[(V, g)] = [(W, g|_W)]$ in C_p^k. But also note that $(W, f|_W) = (W, g|_W)$ and so their germs are equal. It follows that $[(U, f)] = [(V, g)]$.

$(\Longrightarrow)$

Let $W' = U \cap V$. As mentioned before, we know that $[(U, f)] = [(W', f|_{W'})]$ and $[(V, g)] = [(W', g|_{W'})]$. Hence:

$$[(U, f)] = [(V, g)] \Longrightarrow [(W', f|_{W'})] = [(W', g|_{W'})].$$

To simplify our notation, I'll thus assume without loss of generality that $U = V \ (= W')$. And now it suffices to find some open set $W \subseteq U$ containing p such that $f|_W = g|_W$. Fortunately, note that $j_U : C^k(U) \to C_p^k(M)$ given by $j_U(f) = [(U, f)]$